

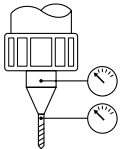
CUTTING CONDITIONS

Drilling | Solid | Cutting conditions

ADO-MICRO 30D

	Mild Steel - Low Carbon Steel SS400 - S10C ~150HB ~500 N/mm ²		Carbon Steel S35C - S50C ~210HB ~710 N/mm ²		Alloy Steel SCM - SCr - sncm 710 ~900 N/mm ²		Alloy Steel SCM - SCr - sncm 710 ~900 N/mm ²		Austenitic Stainless Steel SUS303 - SUS304 SUS316 - SUS316L		Special Alloy Steel SUJ2 - SUS440	
	Vc											
	20~40~60m/min		20~40~60m/min		20~40~60m/min		20~30~40m/min		20~30~70m/min		25~35~45m/min	
Ø	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)
1	12.700	0,01 ~ 0,03	12.700	0,01 ~ 0,03	12.700	0,02 ~ 0,04	9.500	0,02 ~ 0,04	9.500	0,01 ~ 0,03	11.100	0,01 ~ 0,03
1,5	8.500	0,015 ~ 0,045	8.500	0,015 ~ 0,045	8.500	0,03 ~ 0,06	6.400	0,03 ~ 0,06	6.400	0,015 ~ 0,045	7.400	0,015 ~ 0,045
2	6.400	0,02 ~ 0,06	6.400	0,02 ~ 0,06	6.400	0,04 ~ 0,08	4.800	0,04 ~ 0,08	4.800	0,02 ~ 0,06	5.600	0,02 ~ 0,06

	Cast Iron FC250 ~350N/mm ²		Ductile Cast Iron FCD450 - FCD600 400 ~600 N/mm ²		Aluminium Alloy AC4C - ADC		Aluminium A5052 - A7075		Titanium Alloy		Heat Resistant Alloy Inconel 718	
	Vc											
	20~50~60m/min		30~40~50m/min		30~50~70m/min		20~40~60m/min		40~50~60m/min		5~10~15m/min	
Ø	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)	S (min ⁻¹)	F (mm/rev.)
1	15.900	0,02 ~ 0,04	12.700	0,02 ~ 0,04	15.900	0,02 ~ 0,06	12.700	0,01 ~ 0,03	15.900	0,015 ~ 0,025	3.200	0,005 ~ 0,02
1,5	10.600	0,03 ~ 0,06	8.500	0,03 ~ 0,06	10.600	0,03 ~ 0,09	8.500	0,015 ~ 0,045	10.600	0,023 ~ 0,038	2.100	0,008 ~ 0,03
2	8.000	0,04 ~ 0,08	6.400	0,04 ~ 0,08	8.000	0,04 ~ 0,12	6.400	0,02 ~ 0,06	8.000	0,03 ~ 0,05	1.600	0,01 ~ 0,04



1. This cutting condition chart is based on the usage of water-soluble coolant and internal oil supply.
2. Please use quality water-soluble coolant with a dilution factor of approximately 20 times.
3. Please use a precision filter (approximation of 3µm to 5µm) to prevent the oil holes from clogging.
4. Although the recommended coolant pressure is 3 MPa or more, please adjust accordingly if the level of flow volume is unsatisfactory due to the type and concentration of cutting oil used.
5. For accurate mounting, acceptable deflection of the body cylindrical part at the shank end should be less than 0.002µm, as shown in the illustrated figure.
6. For work material with poor chip evacuation characteristic, please perform step drilling as required.
7. From 12D type included, please use a 2D type drill to prepare a pilot hole prior to processing.
8. Please always use the appropriate cutting fluid recommended by the cutting fluid manufacturer in the machining of magnesium alloys. Be cautious with the cutting chips as they are highly flammable and may pose a serious fire risk if not properly handled.